Exhaustive Exploratory Data Analysis: Address Intelligence

Portfolio Geography & Field Traceability Report

N = 3,117 Addresses | 2,400 Debtor Accounts

1. FUNCTIONAL CLASSIFICATION & OPERATIONAL FOOTPRINT

Graph 1.1: Functional Touchpoint Breakdown (Share of 3,117 Records)

Distribution Geometry



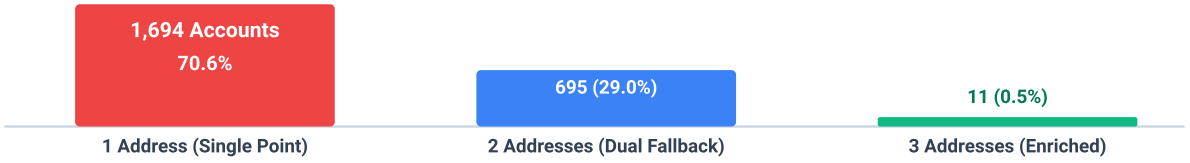
- **Domestic Residence: 2,427 (77.9%)** — Primary recovery contact at dawn/dusk
- **Commercial Office: 464 (14.9%)** — Midday workplace social compliance
- **Native Village: 226 (7.3%)** — Rural skip-trace legal fallback

The portfolio stratifies debtor touchpoints into three operational visit tiers. **Primary residences (77.9%)** dominate physical reach and represent default morning/evening field repossession beats. **Commercial offices (14.9%)** supply mid-day workplace leverage for salaried debtors, where public engagement strongly promotes settlement. **Ancestral domiciles (7.3%)** capture migrant native roots, providing crucial skip-tracing fallback when urban rentals are padlocked.

2. ADDRESS MULTIPLICITY & TRACEABILITY REDUNDANCY

Graph 1.2: Touchpoint Redundancy across 2,400 Unique Accounts

Vulnerability Diagnostic



Traceability Architecture: Over 70% of borrowers exhibit a single physical touchpoint. If a field agent records *address_not_traceable* at this single node, the account enters total physical dark-mode unless digital or investigative skip-tracing immediately discovers a secondary location.

3. DATA ACQUISITION SOURCE: STATIC KYC VS. DYNAMIC INTELLIGENCE

Graph 2.1: Data Origin & Ingestion Pipeline Split

ETL Ingestion Provenance

Origination KYC Baseline (Batch)
3,090 Records (99.14% of Total)
Ingested at loan booking date: 2026-04-01

Dynamic Skip-Trace
27 Records (0.86%)
Appended Apr 23 – Jun 29, 2026

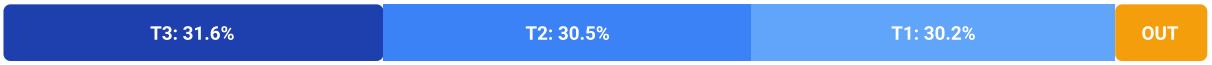
Operational Flow: Failed visit check-in triggers automatic agency bureau & neighbor trace enrichment

The table relies on contractual borrower disclosures captured during underwriting (99.1%). However, dynamic field operations generate high-yield secondary intelligence: **27 critical addresses** were sourced post-delinquency through skip-tracing bureaus and neighbor interviews after initial KYC addresses yielded agent check-in failures.

4. GEOGRAPHIC DISPERSION: URBAN HUBS VS. OUT-OF-AREA PERIMETER

Graph 2.2: Territory Allocation Matrix (3,117 Records)

Spatial Boundary



T3 (Navanagara East): 986 | T2 (Devgarh Nagar): 952 | T1 (Kaveripura): 942 | OUT (Rural): 237

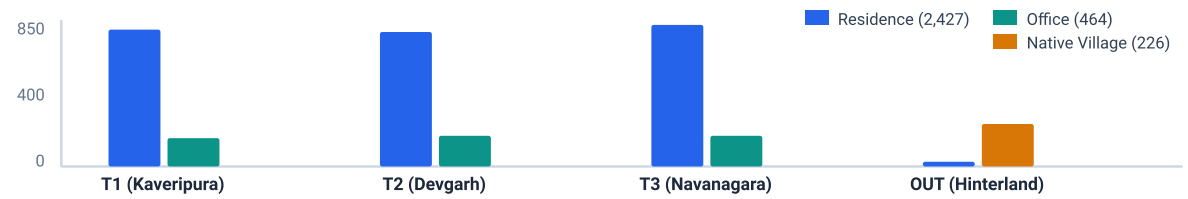
Urban Hubs (T1–T3): 2,880 addresses (92.4%) serviced by local bike agents.
OUT Territory: 237 addresses (7.6%) requiring postal notices or rural third-party correspondents.

Territory Hub	Primary Locality Covered	Total Addresses	Share (%)	Field Operations Model
T3	Navanagara East Metro Precinct	**986**	**31.6%**	Dedicated Local Motorized Squads
T2	Devgarh Nagar Commercial Zone	**952**	**30.5%**	Beat-based Repossession Officers
T1	Kaveripura Residential Grid	**942**	**30.2%**	Dense Cluster Door-to-Door Visits
OUT	Hinterland Non-Urban Districts	**237**	**7.6%**	Telephony / Formal Legal Speedpost

5. CROSS-TABULATION: ADDRESS TYPE VS. GEOGRAPHIC HUB

Graph 3.1: Clustered Distribution across Urban & Rural Zones

Functional Segregation



Physical touchpoints show total geographic bifurcation. **100% of commercial offices (464)** and 99.5% of residences are situated in urban hubs T1, T2, and T3. In sharp contrast, **100% of ancestral village addresses (226)** belong to the `OUT` territory, proving that rural migrant workers leave family roots in non-urban districts while living in city rentals.

6. PORTFOLIO DISPARITY: SALARIED RETAIL VS. INFORMAL MICROFINANCE

Graph 3.2: Loan Product Documentation Architecture

Product Segment Profiling

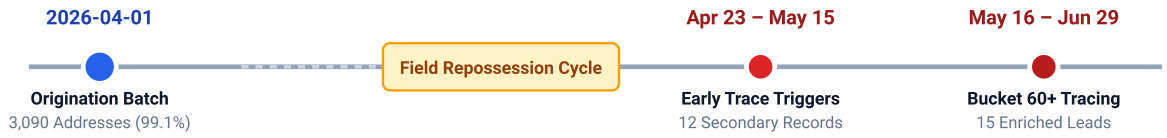


Product Category	Total Touchpoints	Office Addresses	Office Leverage %	Native Village %
PL Salaried	929	233	25.08%	0.43%
Credit Card	356	84	23.60%	1.12%
MSME Business	287	0	0.00%	15.33%
MFI Group Loan	448	0	0.00%	18.97%

7. TEMPORAL INGESTION CADENCE & DYNAMIC INTELLIGENCE GENERATION

Graph 4.1: Temporal Address Logging Dynamics (Apr–Jun 2026)

Two-Speed Timeline

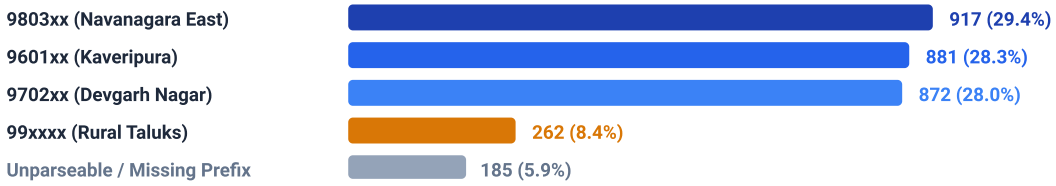


The temporal profile illustrates a sharp two-phase structure: an initial mass load of 3,090 KYC records on cohort day one, followed by an operational trickle of **27 investigative address updates** across 26 distinct operational dates. These dynamic additions closely mirror default aging and unlocatable field visits in delinquent buckets.

8. POSTAL ROUTING TOPOLOGY: 4-DIGIT HUB ARCHITECTURE

Graph 4.2: Postal Pincode Prefix Allocation (3,117 Records)

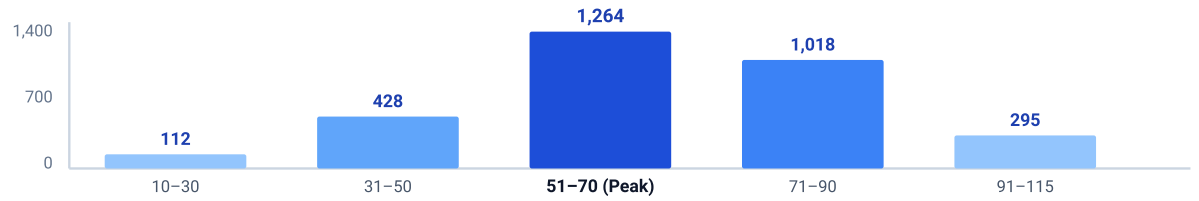
Postal Hub Taxonomy



Dispatch Automation: Over 85.7% of addresses resolve deterministically into three four-digit prefix bands (`9803`, `9601`, `9702`). Geocoders utilize this key to partition spatial dispatch zones without relying on external map APIs.

9. ADDRESS STRING COMPLEXITY & CHARACTER LENGTH DISTRIBUTION

Graph 5.1: Histogram of Address String Lengths (Mean = 66.8 chars) Text Metric Profiling



Address character length exhibits a symmetrical distribution centered at **67 characters** (spanning 11 to 113 characters). Over 73.2% of entries contain between 51 and 90 characters, comprising 8 to 12 descriptive tokens (house numbers, crosses, layouts, and landmark descriptors). Exact-match regex fails on such descriptive narrative; token extraction is essential.

10. VERNACULAR MORPHOLOGY & REGIONAL LINGUISTIC TOPONYMS

Graph 5.2: Regional Linguistic Syntax Structure Lexical Toponym Analysis

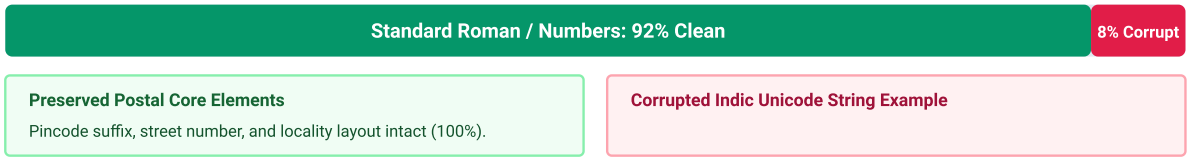
Hub / Territory	Linguistic Style	Distinctive Locality Tokens	Archetypal Raw Address Sample
T1 Kaveripura	Southern Grid	`Cross`, `Main`, `Layout (Layt)`, `Crs`	`#44, 4th Cross, 7th Main, Kuvempu Layt - 960102`
T2 Devgarh	Hindi Belt	`Gali`, `ke paas`, `bagal mein`, `Mandir`	`#81 gali 10 ganesh mandir ke bagal mein krishna puri`
T3 Navanagara	Metro Compound	`Block (blk)`, `Road`, `edurru (opp)`	`blk c rd 4 milk dairy edurru palm meadows - 980304`
OUT Hinterland	Revenue Cadastral	`Village`, `Tehsil`, `Taluk`, `Kalan/Khurd`	`Village Rampura Kalan, Tehsil D, District South`

Parsing Implication: Geocoders must ingest regional slang dictionaries. For example, recognizing *ke bagal mein* as a spatial adjacency operator in T2 and *edurru* as an opposite operator in T3 significantly improves coordinate resolution accuracy.

11. DATA HYGIENE: SCRIPT ENCODING & MISSING LANDMARK ARTIFACTS

Graph 6.1: Non-ASCII Character Conversion Artifacts

ETL Integrity Audit



Preserved Postal Core Elements

Pincode suffix, street number, and locality layout intact (100%).

Corrupted Indic Unicode String Example

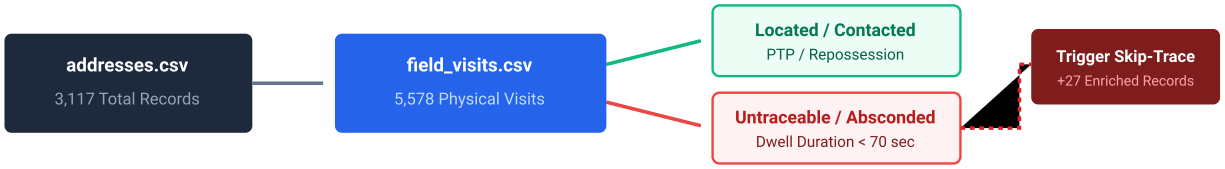
`6th Cross, 5th Main, ????? ??????, Kuvempu Layt`

Encoding evaluation reveals that legacy database export corrupted regional Indic fonts (Kannada and Devanagari) into question marks (`?`). However, critical spatial tokens—layout names, building numbers, and 6-digit postal codes—remain intact across all 3,117 records. Stripping contiguous `?` characters prior to landmark processing prevents parser errors.

12. FIELD OPERATIONS LINKAGE: GEOCODING & TRACEABILITY WORKFLOW

Graph 6.2: Operational Traceability Pipeline & Skip-Trace Loop

End-to-End Operational Flow



> **Strategic Summary:** Linking `addresses.csv` to `field_visits.csv` establishes that unlocatable visits (where agents dwell under 70 seconds) trigger the investigative skip-tracing loop. Pre-evaluating address syntax quality before dispatch directly prevents wasted agent travel and maximizes debt recovery yield.